

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination May 2016

PH212 Physical Pharmaceutics

Date: 11.05.2016, Wednesday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A (i) What do you mean by Micromeritics? Discuss the applications of Micromeritics in Pharmacy. 05
- (ii) Write a note on polymorphism. 05
- B (i) Explain in brief: (a) DSC (b) XRD 05
- (ii) Explain the following terms with suitable examples: 05
- (a) Sublimation (b) Eutectic Mixture
- C (i) Enumerate various methods used for determination of particle size. 05
- Discuss any one method in detail.
- (ii) Which are the derived properties of powders? Discuss the 'Angle of repose' with relation to powder flow. 05

Q 2 Attempt any TWO of the following

- A (i) Define and classify Colloidal dispersions. 05
- (ii) Explain the applications of colloids in Pharmacy. 05
- B (i) Discuss the factors influencing settling of particles in suspension. 05
- (ii) Differentiate between Flocculated and Deflocculated suspension. 05
- C (i) Define and classify Emulsions. How do you prepare primary emulsion? 05
- Explain any two identification tests for emulsions.
- (ii) Write a note on Physical stability of Emulsion. 05

SECTION- II

Q 3 Attempt any **TWO** of the following

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|---|---|----|
| A | Enlist different methods for the measurement of surface and interfacial tensions. Explain any one method in detail. | 10 |
| B | (i) Explain in brief: Spreading co-efficient. | 07 |
| | (ii) Define: (a) Buffer capacity (b) Isotonic solution (c) Zeta potential | 03 |
| C | Enlist various pharmaceutical and physiological buffers. Explain any two pharmaceutical Buffers in detail. | 10 |

Q 4 Attempt any **TWO** of the following

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|---|---|----|
| A | Write a detailed note on application of complexes in pharmacy. | 10 |
| B | (i) Explain the principle and application of capillary viscometer. | 05 |
| | (ii) Write a note on: Monomolecular inclusion complex. | 05 |
| C | Define 'Non-Newtonian Flow'. Explain different types of 'Non-Newtonian Flow' in detail. | 10 |
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